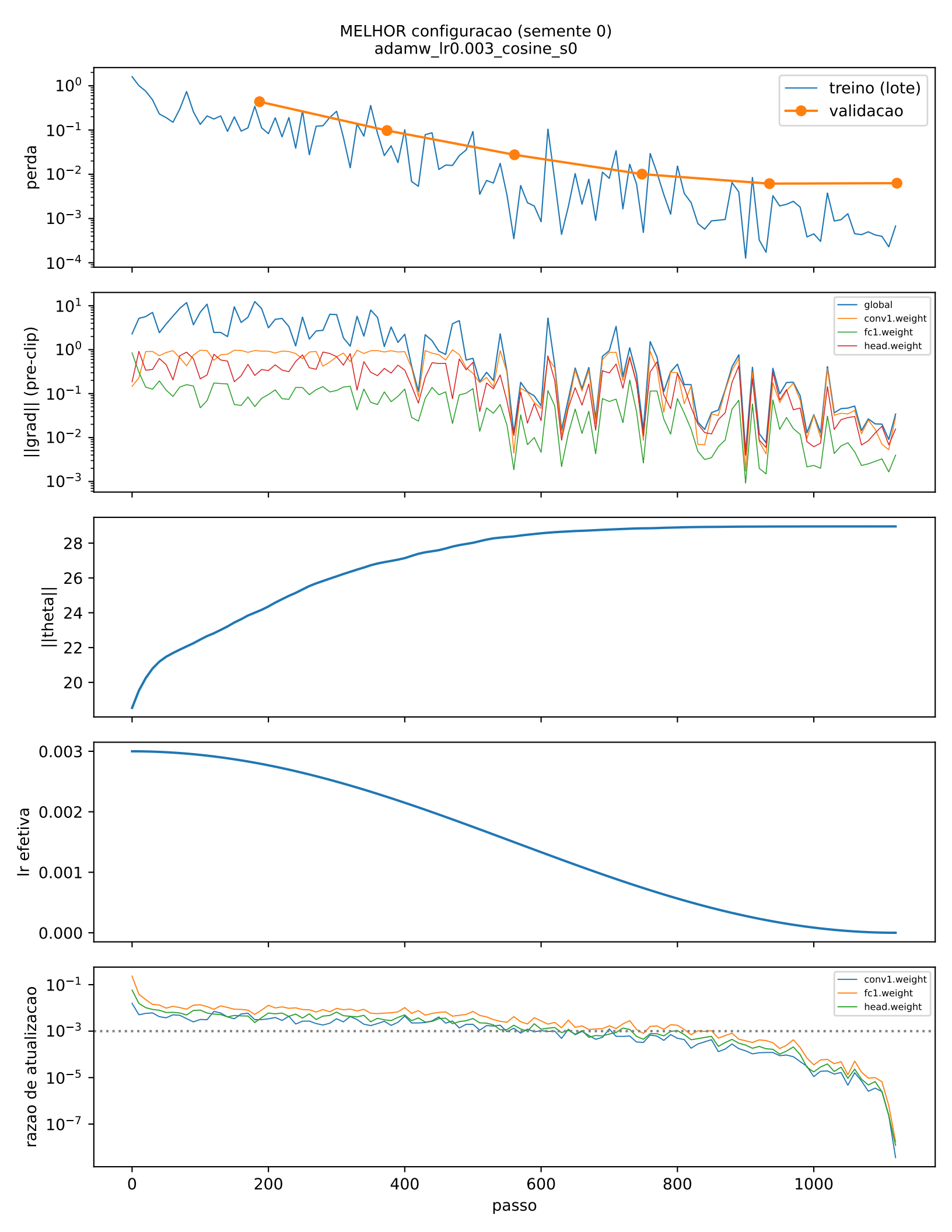
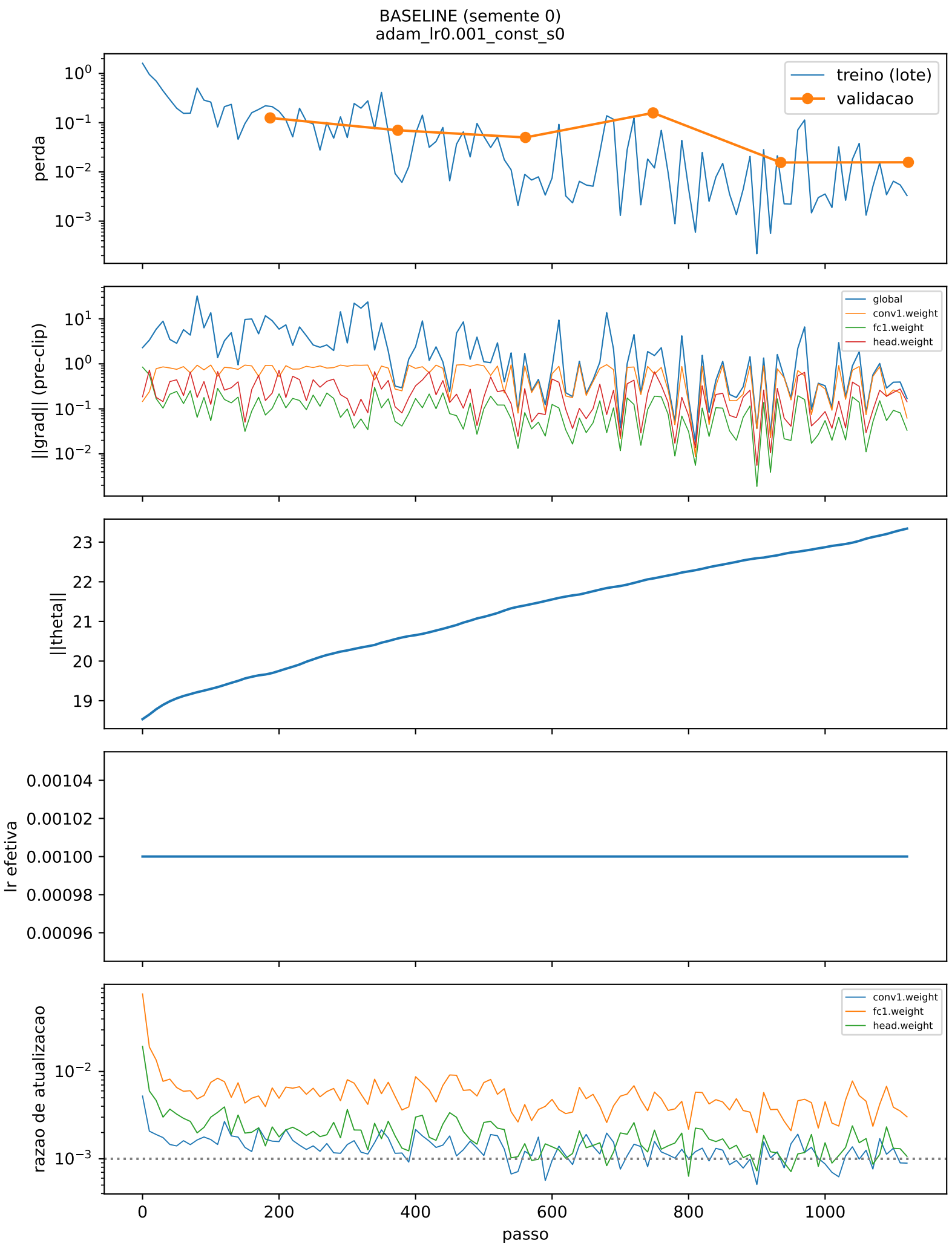
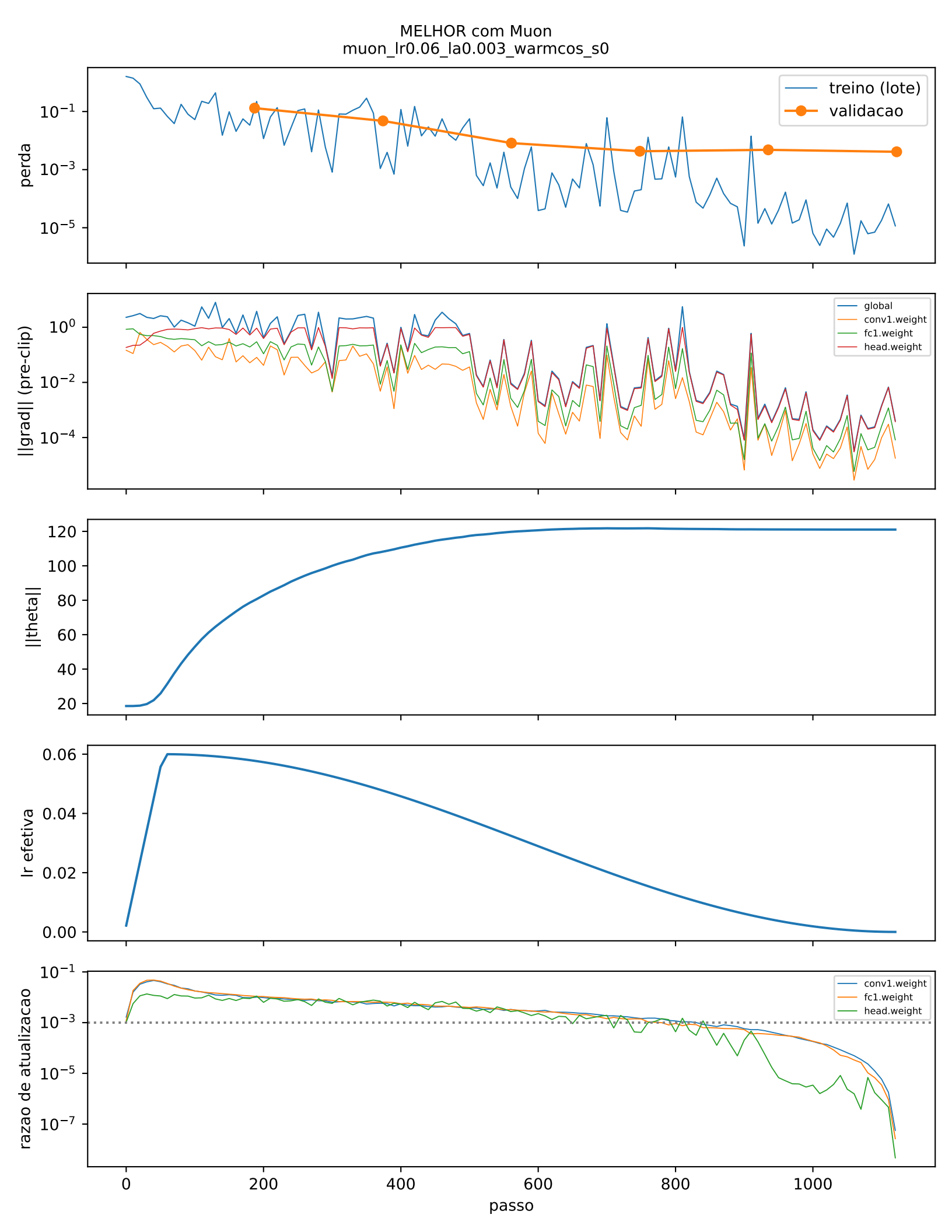
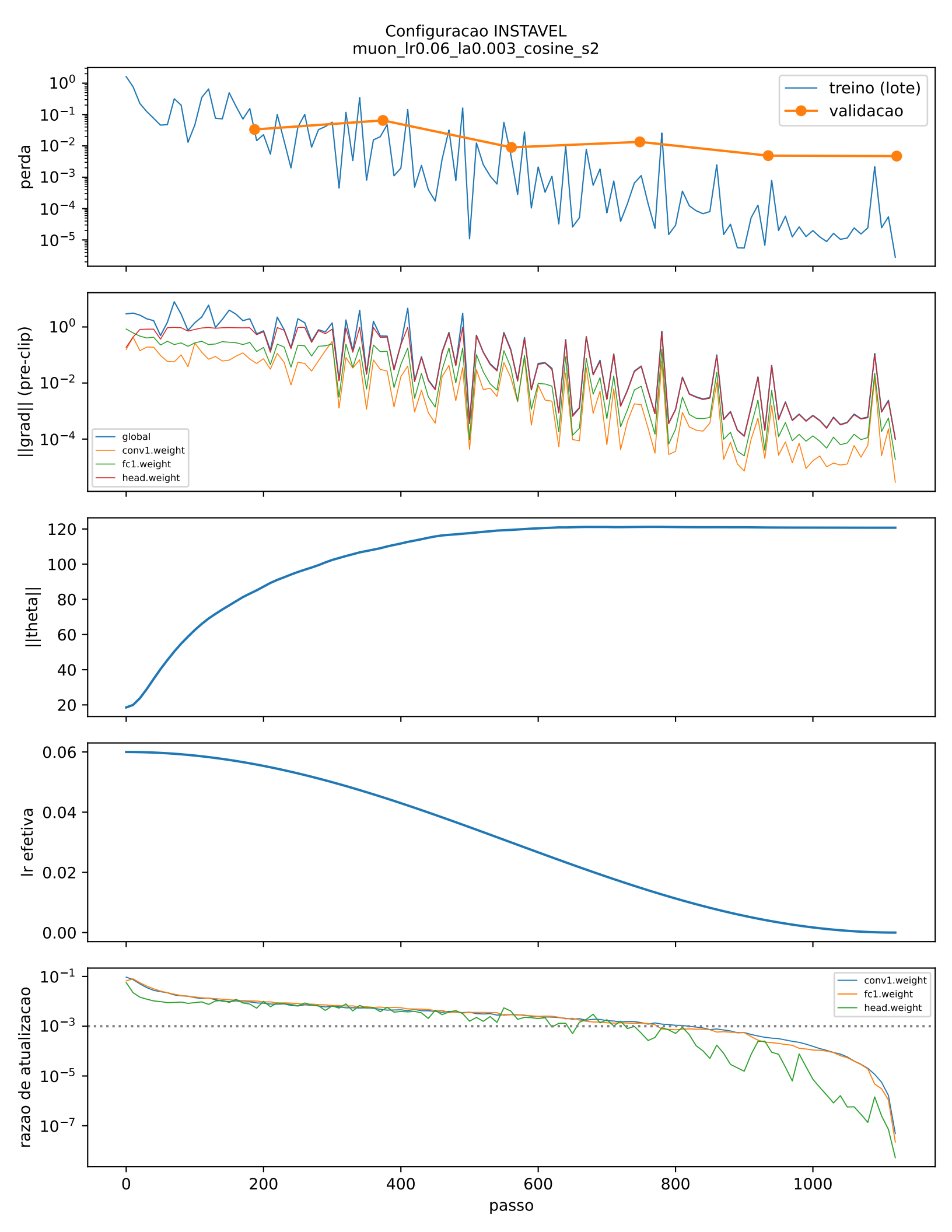
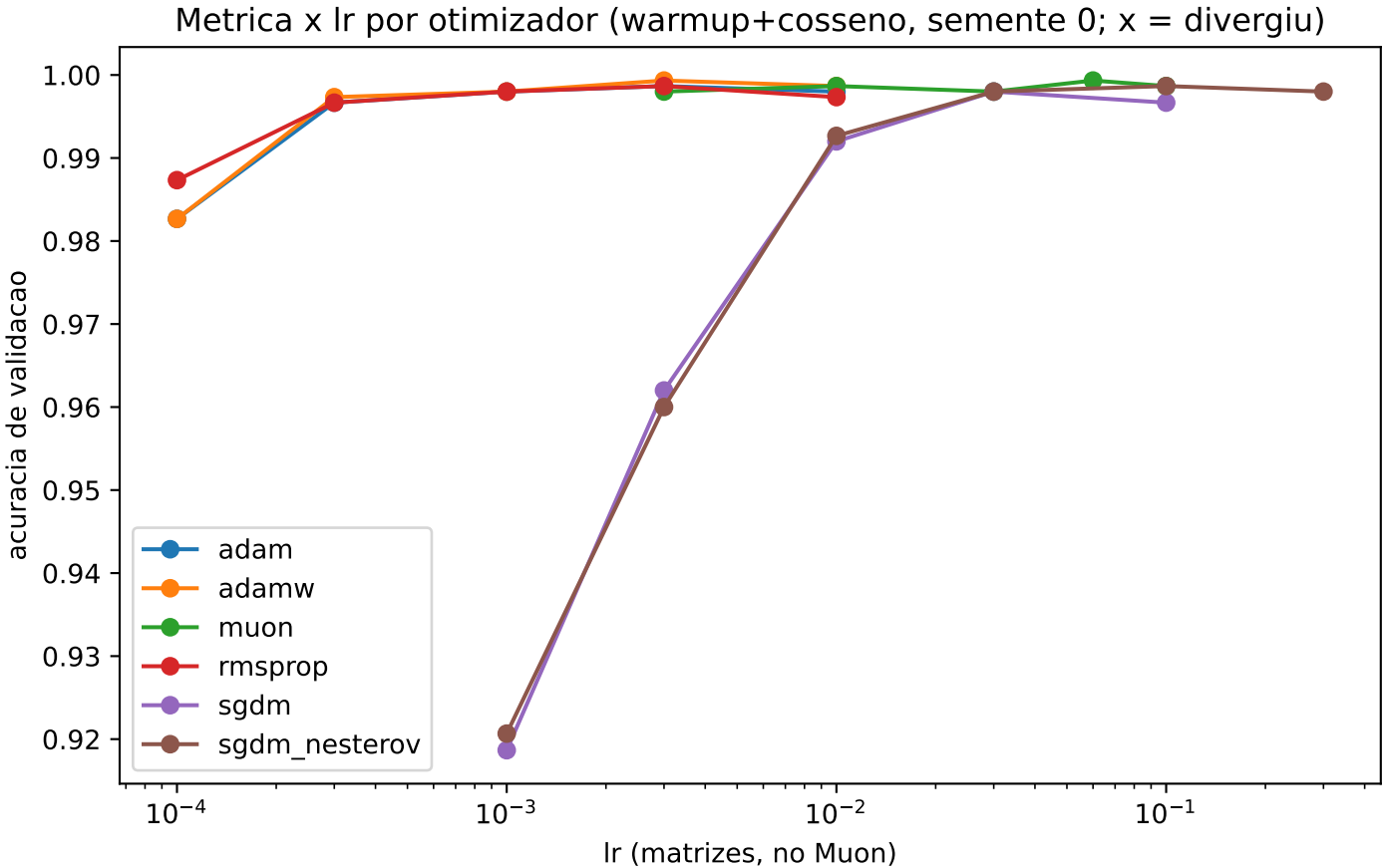

MELHOR configuracao (semente 0)
adamw_lr0.003_cosine_s0
perda
treino (lote)
validacao
||grad|| (pre-clip)
global
conv1.weight
fc1.weight
head.weight
||theta||
lr efetiva
razao de atualizacao
conv1.weight
fc1.weight
head.weight
passo


BASELINE (semente 0)
adam_lr0.001_const_s0
perda
treino (lote)
validacao
||grad|| (pre-clip)
global
conv1.weight
fc1.weight
head.weight
||theta||
lr efetiva
razao de atualizacao
conv1.weight
fc1.weight
head.weight
passo


MELHOR com Muon
muon_lr0.06_la0.003_warmcos_s0
perda
treino (lote)
validacao
||grad|| (pre-clip)
global
conv1.weight
fc1.weight
head.weight
||theta||
lr efetiva
razao de atualizacao
conv1.weight
fc1.weight
head.weight
passo


Configuracao INSTAVEL
muon_lr0.06_la0.003_cosine_s2
perda
treino (lote)
validacao
||grad|| (pre-clip)
global
conv1.weight
fc1.weight
head.weight
||theta||
lr efetiva
razao de atualizacao
conv1.weight
fc1.weight
head.weight
passo


Metrica x lr por otimizador (warmup+cosseno, semente 0; x = divergiu)
acuracia de validacao
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
adam
adamw
muon
rmsprop
sgdm
sgdm_nesterov
10^{-4}
10^{-3}
10^{-2}
10^{-1}
lr (matrizes, no Muon)